CHE-251 END TERM REPORT

Title: Material Balances for Ammonia Production via Haber Process with Recycle and Purge

Course Code: CHE 251 – Process Calculations

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End Term Report: Ammonia Production via Haber Process

Abstract

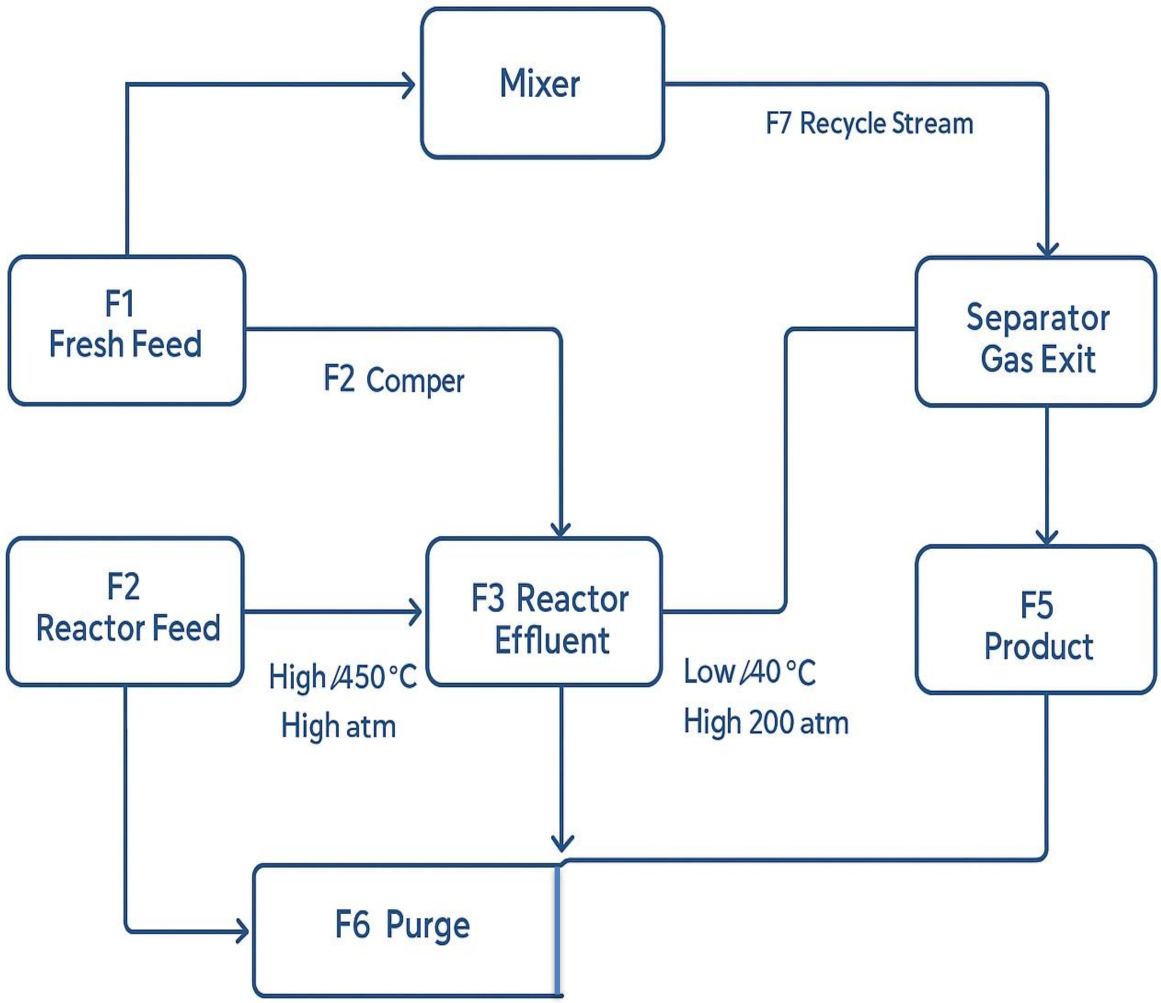
The Haber process is an industrially significant reaction used to synthesize ammonia (NH_3) from nitrogen (N_2) and hydrogen (H_2) gases. Due to the inherent limited single-pass conversion (typically 15–35%) imposed by equilibrium constraints, a complex system incorporating recycle and purge streams is essential to enhance overall efficiency and achieve high overall yields. This report presents the detailed material balance calculations for such a system, focusing on the critical effect of the inert component (Argon) and the necessity of the purge stream to prevent its accumulation. A comprehensive process flow diagram (PFD), step-by-step unit operation analysis, and sensitivity analysis are presented. Findings confirm that an optimized operation requires balancing reactor performance (high conversion) against process stability (low inert concentration), which is achieved with a conversion of 25–30% and a corresponding low purge ratio.

1.0 Introduction

Ammonia is a cornerstone industrial chemical, primarily used in fertilizer manufacturing and for the production of explosives. Its production is dominated by the Haber-Bosch process, which involves the highly exothermic and reversible gas-phase reaction:

$$N_2 + 3H_2 \rightleftharpoons 2NH_3$$

The thermodynamic equilibrium of the Haber reaction favors ammonia formation at low temperatures. However, the reaction rate is too slow at low temperatures, necessitating high temperatures (typically 400–500°C) and extremely high pressures (150–250 atm) to achieve a reasonable reaction speed with an iron catalyst. This operating compromise results in a low single-pass conversion, often limited to 20–30%. To recover the unreacted nitrogen and hydrogen, a system involving a separator and



recycle loop is employed. The purge stream is introduced as a critical mechanism to maintain process stability by controlling the concentration of inert gases (primarily Argon).

2.0 Process Description and Flow Diagram

The process flow diagram (PFD) below illustrates each stream and its function:

Table 1: Stream Descriptions and Conditions

Stream	Description	Components	State Variables
F1	Fresh Feed	N_2, H_2, Ar	T=25°C, P=1 bar
F7 (FR)	Recycle Stream	N_2, H_2, Ar	T≈40°C, P=200 atm
F2	Reactor Feed	N_2, H_2, Ar	T=450°C, P=200 atm
F3	Reactor Effluent	N_2, H_2, NH_3, Ar	T≈450°C, P=200 atm
F4	Separator Gas Exit	N_2, H_2, Ar	T=40°C, P=200 atm
F5 (FP)	Product Stream	NH_3	T≈40°C (Liquid)
F6 (FPurge)	Purge Stream	N_2, H_2, Ar	T≈40°C, P=200 atm

Table 2: Basis of Calculation and Design Assumptions

Parameter	Value	Basis/Source
Overall Basis	100 kmol/hr	Total Fresh Feed Flow (F1)
Stoichiometry	H_2:N_2 = 3:1	Reaction requirement ($N_2 + 3H_2 \rightarrow 2NH_3$)
Inert Component	Ar = 0.5 mol% (0.5 kmol/hr)	In Fresh Feed
Reactor Conversion (X)	15% – 35%	Range for sensitivity analysis

Separator Performance	100% NH_3 removal	Ideal separation assumption
Stability Criterion	Based on Argon balance	Process stability requirement
Pressure Drop	Negligible	Assumed across non-reactive units.

3.0 Step-by-Step Unit Operation Analysis

This detailed analysis validates the methodology used to compute material balances for each unit operation in the ammonia process loop.

3.1 Mixer

Purpose: Combines the low-flow Fresh Feed (F1) and the high-flow Recycle Stream (FR) to form the combined Reactor Feed (F2). Calculation: Total and component mass conservation: F2 = F1 + FR. Temperature T2 is determined by an adiabatic energy balance since no reaction occurs.

3.2 Reactor (Ammonia Converter)

Purpose: Drives the exothermic synthesis reaction at high pressure and temperature in the presence of a catalyst. Calculation: The material balance is defined by the extent of reaction (ξ) based on N_2 single-pass conversion X. The flow of NH_3 formed is $2 \times (X \times F_{2},N_{2})$. Isothermal operation is often assumed for the conversion reactor.

3.3 Separator/Condenser

Purpose: Cools the reactor effluent (F3) to condense and remove the product NH_3. Calculation: Assumed ideal split — 100% of NH_3 is removed as product (F5), while remaining gases (F4) are retained for recycle and purge.

3.4 Stream Splitter and Purge

Purpose: Divides the unreacted gas stream (F4) into the Recycle (FR) and Purge (FPurge) streams. Calculation: The flow rate of the purge is determined by the steady-state overall inert balance. The compositions of F4, F6, and F7 are assumed identical.

4.0 Material Balance Derivations

The key to solving this process is using overall balances (enclosing the entire process, relating F1, F5, F6) combined with the local balances at the reactor.

4.1 Overall Argon (Inert) Balance

At steady state, the inert material entering the system (F_1,Ar) must equal the inert material leaving through the purge (F_6,Ar):

F_1,Ar = F_6,Ar

$F_6 \times$ y_Ar = F_1,Ar → F_6 = F_1,Ar / y_Ar [Equation 1: Purge Flow]

4.2 Overall Nitrogen Balance (Yield Equation)

N_2 Balance: N_2 in Fresh Feed = N_2 in Purge + N_2 Consumed.
F_1,N_2 = F_6,N_2 + ½ F_5
Substituting y_N_2 = ¼ (1 - y_Ar) and F_6 = F_1,Ar / y_Ar gives:

F_5 = 2 × [F_1,N_2 - (F_1,Ar / 4) × ((1 - y_Ar) / y_Ar)] [Equation 2: Overall Yield]

5.0 Worked Example: Base Case Calculation

Base Case: X = 25% conversion, F_1 = 100 kmol/hr (with F_1,Ar = 0.5 kmol/hr), and stability criterion y_Ar = 0.04 (4 mol% in the loop).

Purge Flow: $F_6 = 0.5 / 0.04 = 12.5$ kmol/hr

Product Yield: $F_5 = 2 \times [24.75 - (0.5 / 4) \times ((1 - 0.04) / 0.04)] = 43.5$ kmol/hr

Reactor Feed N_2: $F_{2,N_2} = N_2$ Consumed / X = 21.75 / 0.25 = 87.0 kmol/hr

Total Reactor Feed: $F_2 = 87.0 + 261.0 + 12.5 = 360.5$ kmol/hr

Table 3: Detailed Material Balance (Base Case)

Component (kmol/hr)	F1	F2	F3	F4	F6	F5
N_2	24.75	87.0	65.25	65.25	2.76	—
H_2	74.75	261.0	195.75	195.75	8.28	—
NH_3	—	—	43.5	—	—	43.5
Ar	0.5	12.5	12.5	12.5	0.5	—
Total Flow	100.0	360.5	317.0	273.5	11.5	43.5
y_Ar (%)	0.5	3.47	3.94	4.57	4.33	—

6.0 Results and Discussion

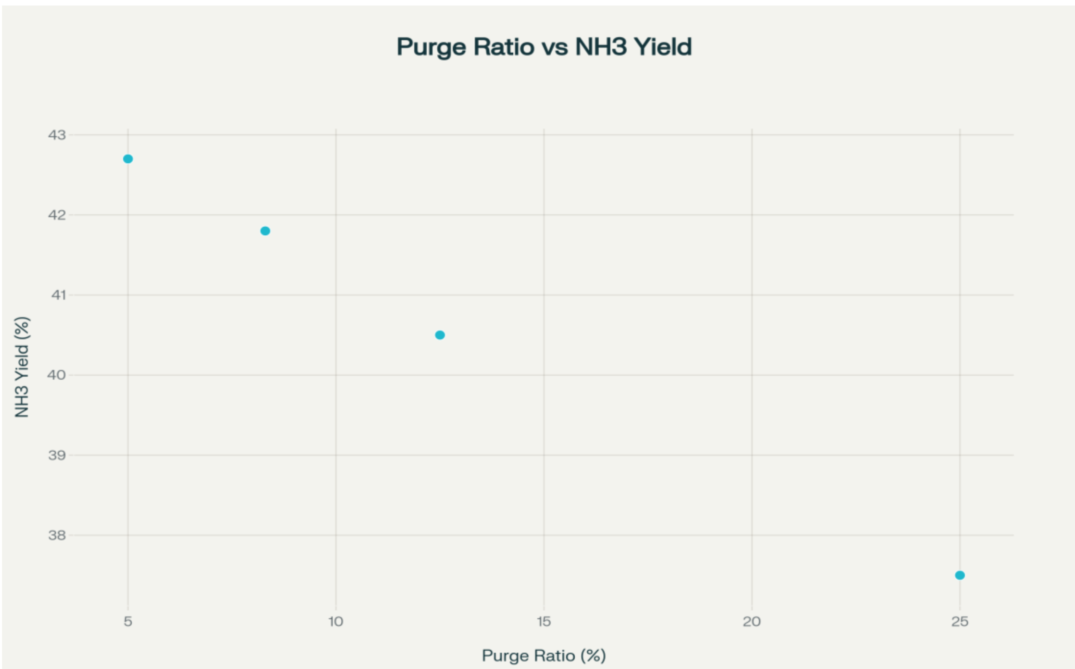
The sensitivity analysis confirms the strong interdependence between single-pass conversion, purge ratio, and overall yield.

Table 4: Sensitivity of Overall Yield to Purge Ratio (X = 25%)

Target y_Ar in F4 (%)	Calculated Purge Ratio (F6/F1)	Overall NH_3 Yield (%)

2%	25.0%	37.5
4%	12.5%	40.5
6%	8.3%	41.8
10%	5.0%	42.7

Figure 2: Effect of Purge Ratio on Overall Ammonia Yield



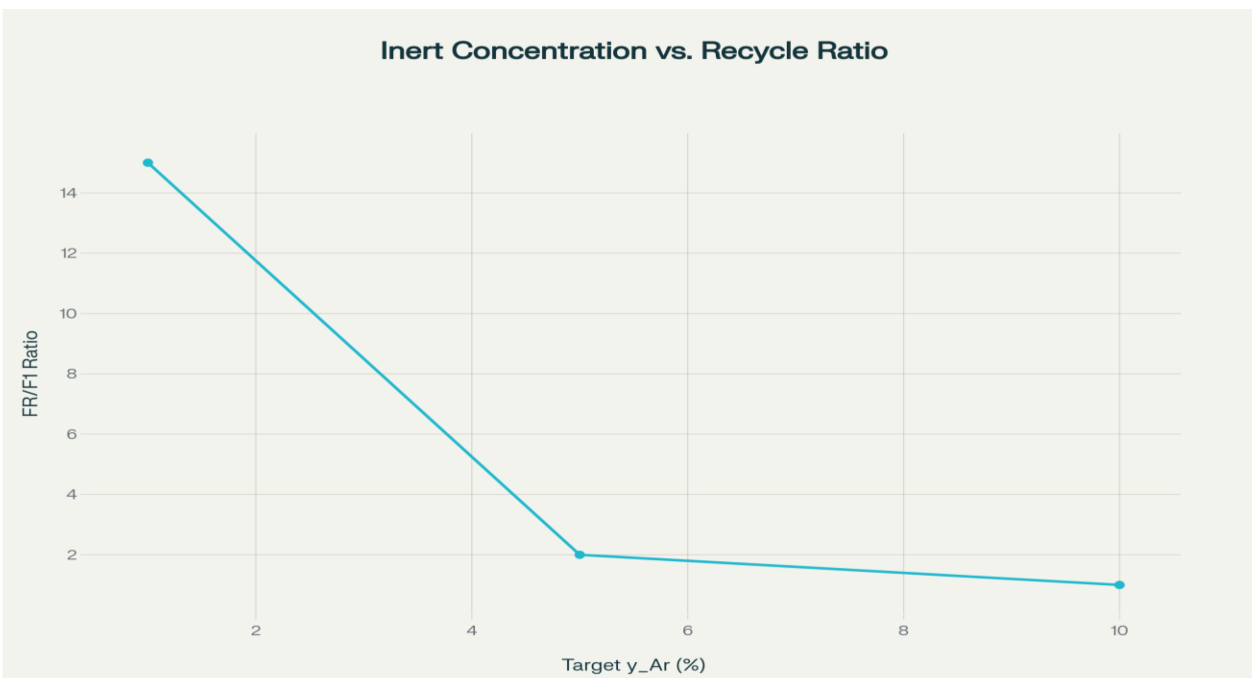
6.2 Optimization and Industrial Comparison

Industrial optimization criteria primarily address catalyst deactivation and reactor efficiency. High inert concentrations reduce the partial pressure of reactants, lowering the effective kinetic driving force. Therefore, industrial systems maintain inert concentrations below 8–10% to ensure stability. Operating with 25–30% conversion and purge flows maintaining 4–6% inert concentration achieves the best economic balance.

Table 5: Stability Analysis - Inert Concentration vs. Recycle Flow

Target y_Ar in F4 (%)	Required Purge Flow (F6) (kmol/hr)	Recycle Flow (FR) (kmol/hr)	FR / F1 Ratio
1%	50	1500	15:1
5%	10	200	2:1
10%	5	100	1:1

Figure 3: Relationship between Inert Concentration and Required Recycle Ratio on the next page



7.0 Conclusion

This end-term report presented a comprehensive material balance analysis for the Haber process with recycle and purge streams. The inclusion of the Argon balance demonstrated its controlling influence over the purge flow rate. Sensitivity analysis confirmed that a low purge is desirable for yield but a minimum purge is required for steady operation. Operating at 25–30% conversion and 4–6% inert concentration aligns with industrial ammonia production standards.

8.0 DWSIM Simulation of the Process

A DWSIM Simulation for the process where I assumed the single pass conversion to 25% and the purge ratio to be 5%. The link for the same is given as follows :

https://drive.google.com/file/d/10rSGpL03VinQe6VkzOquE_g_2n2ij3c3/view?usp=sharing

(For viewing the simulation download the above drive file, then open with DWSIM to view it).

9.0 Summary of Learnings / Reflection

The key learning from this project is understanding that in a recycle system with an inert component, overall yield and flow are governed not just by single-pass conversion but by the inert balance constraint. The concept of calculating purge flow based on a stability constraint rather than arbitrarily setting it mirrors real-world process design p